The Real Effects of Valuation Mistakes: Estimates from Mergers and Acquisitions

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Motivation

Pricing effects of behavioral biases can be arbitraged away, but real decisions are more difficult to arbitrage away

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We focus on a well-documented bias: thinking in $ rather than in %

- Nominal illusion
- Denominator neglect
- *"It's easier for a $4 stock to go to $8 than for a $40 stock to go to $80"* (Fidelity's low-price stock fund manager cited in Birru & Wang 2016)
- Non-proportional thinking: *"investors partially think about stock price changes in dollar rather than percentage units"* (Shue & Townsend 2021)

Applying this logic to M&A transactions

Firm A
$1 bn market cap
100M shares outstanding
P = $10
Offer premium = 40% = $4

Firm B
$1 bn market cap
50M shares outstanding
P = $20
Offer premium = 40% = $8

Biased investor thinks an acceptable premium is $6
⇒ $4 premium for firm A is perceived as low, $8 premium for firm B is perceived as high

If enough investors are subject to this bias, offer premiums and acquisitions are distorted

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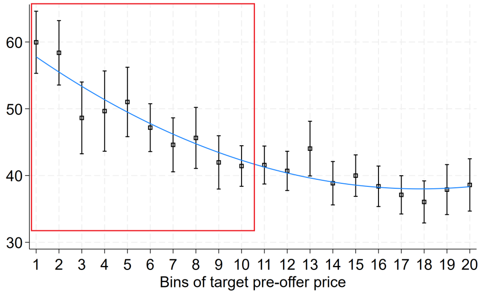
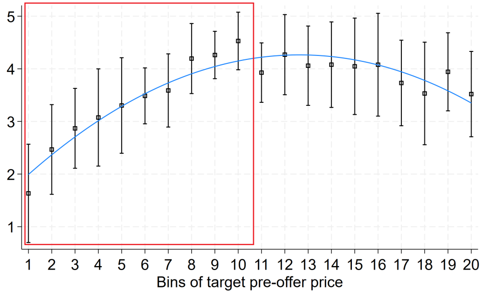
Panel A: Offer Premium (%)



Panel B: Acquisition Probability (%)



Controls: target market cap (20 bins), year FE

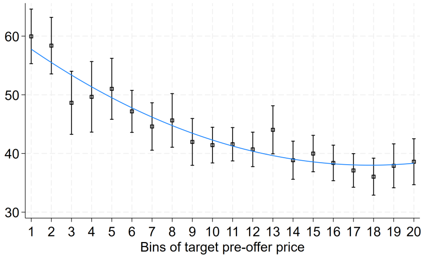
Results

Low-price targets have high premium and low acquisition probability

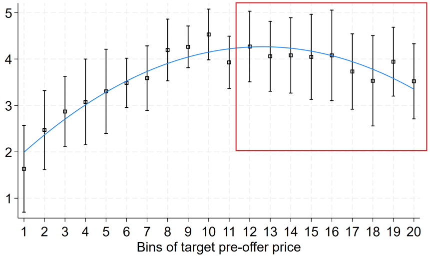
1. Robust to tighter controls
   - Target characteristics: M/B, 52-week high, cash, PPE, leverage, profitability, industry FE
   - Deal characteristics: tender, cash payment, hostile, poison pill, public acquirer, horizontal, toehold
2. More pronounced when target shareholders are less sophisticated
   - Proxy: one minus % institutional ownership

Results

Panel A: Offer Premium (%)



Panel B: Acquisition Probability (%)



Acquisition probability is (somewhat) hump-shaped, i.e., low for high-price firms

- Consistent with non-proportional thinking by the acquirer $\Rightarrow$ unwilling to pay high premium
- Abstract from acquirer bias for this presentation (see paper for two-sided bias)

Questions

1. What does the premium effect imply for value split between target shareholders and acquirer shareholders?

2. What does the acquisition probability effect imply for total value creation through M&As?

$\Rightarrow$ A model to quantify these effects

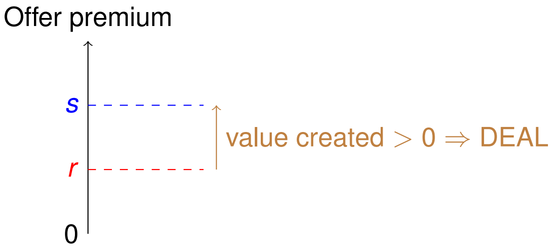
Model

Target with price P is matched with a bidder

- Bidder's valuation for the target: $(1+s)P$
- Target shareholders' reserve price: $(1+r)P$
- $s > 0$ and $r > 0$ drawn from exponential distributions

Rationale benchmark

- Deal takes place iff $s > r$
- Offer premium r



Non-Proportional Thinking

Non-proportional thinking (NPT)

- Target shareholders require <u>fixed</u> dollar premium C instead of proportional rP
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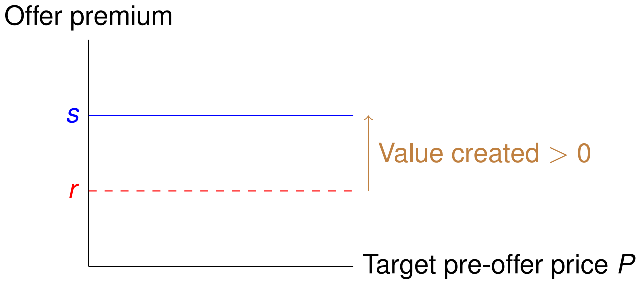
Assume mngt is unbiased and sale requires both mngt's and shareholders' approvals

$\Rightarrow$ Target sells iff premium $> \max\left\{r, r^{1-\theta}\left(\frac{C}{P}\right)^{\theta}\right\}$

Non-Proportional Thinking



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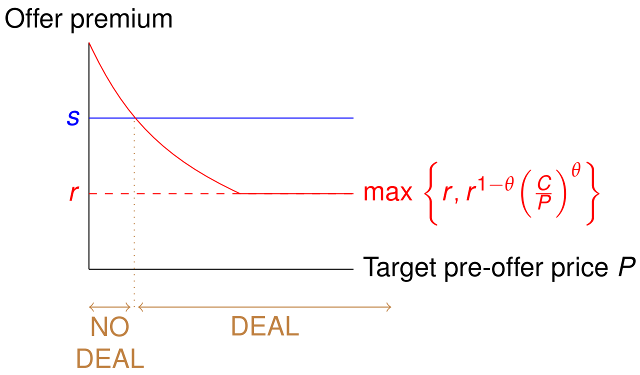

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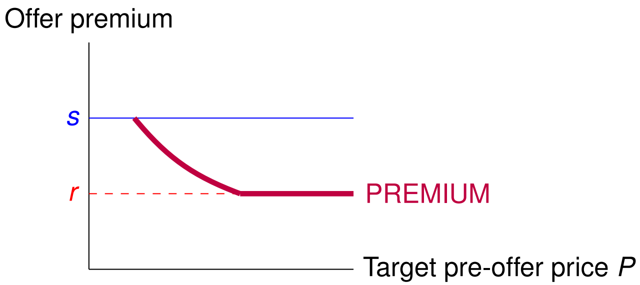


Identification

Parameters to estimate:

1. Probability of becoming a target (q)
2. Mean synergies ($E[s] = 1/\mu$)
3. Mean reservation price ($E[r] = 1/\lambda$)
4. Intensity of NPT (θ)
5. Fixed $ premium used by biased shareholders (C)

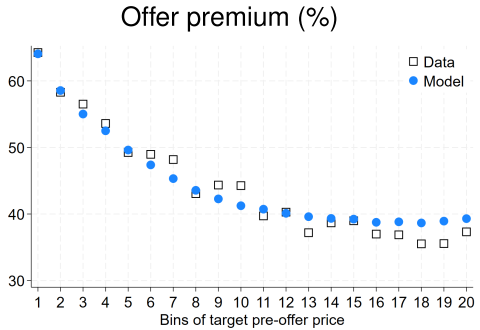
Moments:

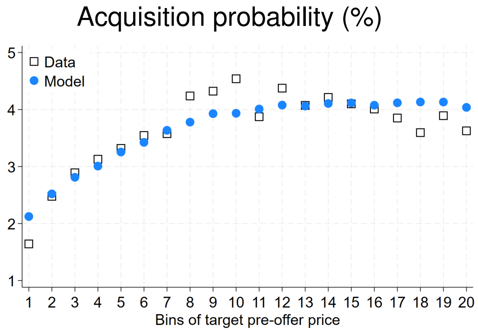
1. Deal frequency (identifies q)
2. Average premium (identifies λ, μ)
3. Relation between deal probability and P (identifies (identifies θ, μ)
4. Relation between premium and P (identifies θ)
5. Assume zero average bias (identifies C)

GMM Estimates and Fit

θ	C	λ	μ	q
.16***	13***	1.5***	.78**	.056***
(.045)	(.75)	(.32)	(.32)	(.013)

Shareholders put weight 0.16 on the biased premium and 0.84 on the rational premium





Impact of Bias

	Fitted model (1)	Counterfactual with $\theta = 0$ (2)	Impact of bias (1) minus (2)
Acquisition probability (%)	3.66	3.74	-.073
Average premium (%)	44.3	42.6	1.7
Expected value creation (%)	6.25	6.33	-.078
for acquirer	4.63	4.74	-.11
for target	1.62	1.59	.031

1. Acquisition probability decreases by 2% (0.073 pp)
2. Value created by M&As decreases by 1.2% (0.078 pp)
   - Smaller effect than on acquisition probability because marginal deals create less value
3. Value is redistributed from acquirer shareholders to target shareholders
   - Lower acquisition probability is more than offset by higher premium

Summary

We study how non-proportional thinking by investors affects M&A transactions

1. M&A premia are a decreasing function of the target's pre-offer price

2. Low-price firms are less likely to be acquired

Reduced-form evidence

Structural estimates are economically plausible

The bias reduces the frequency of M&As by about 6% and the value created by the M&A market by about 1.2%